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Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

I have received your dataset: `sicor_municipal_contracts.csv`. I'll start by loading and inspecting the data to understand its structure and key characteristics. Then, you can provide a specific analytical question, and I'll extract insights accordingly.

Let me first check the dataset's contents.

The dataset contains **231,030 rows** and **19 columns**, with numerical data covering state and municipality codes, funding and investment quantities/values, economic activities, and contract values.

Key observations:

- The **ContractTotalValue** column likely represents the total value of each contract.
- Funding, investment, commercialization, and industrialization quantities and values are included, indicating financial flows in different economic activities.
- **StateCode** and **MunicipalityCode** suggest location-based analysis is possible.
- **MonthIssued** allows for temporal analysis.

Now, what specific analytical question would you like me to answer using this dataset? 🚀 [?]

institutions will have the largest participation in granting commercialization investments in upcoming periods, based on their historical resource distribution and other available factors? Define three options to address this issue and select the best one.

Great question! Predicting which financial institutions will have the **largest participation in commercialization investments** requires a structured approach. Here are three possible methods:

Option 1: Time Series Forecasting (ARIMA, Prophet, LSTM)

🔥 Approach:

- Use historical **CommercializationValue** data per **FinancialInstitutionCNPJ** over time.
- Apply **time series models** (ARIMA, Prophet, or LSTM) to forecast future investment allocations.

📊 Pros:

- ✓ Captures trends and seasonality.
- ✓ Effective for institutions with long-term data.

⚠️ Cons:

- ✗ Needs enough historical data points per institution.
- ✗ Doesn't directly consider external factors like economic shifts.

Option 2: Machine Learning Classification (Random Forest, XGBoost, Logistic Regression)

🔥 Approach:

- Define **high vs. low participation** institutions based on past commercialization investments.
- Train a **classification model** using features like **ResourceSourceCode**, **EconomicActivity**, **FundingValue**, etc.

- Predict which institutions are likely to be high contributors in the future.

Pros:

- ✓ Considers multiple influencing factors.
- ✓ Identifies key predictors for commercialization investments.

Cons:

- ✗ Requires a well-balanced dataset.
- ✗ May be less accurate for small or new institutions.

Option 3: Regression-Based Ranking (Multiple Linear Regression, Gradient Boosting)

Approach:

- Use **CommercializationValue** as the target variable.
- Train a **regression model** to estimate future values for each financial institution.
- Rank institutions by predicted investment amounts.

Pros:

- ✓ Can incorporate multiple influencing factors.

Cons:

- ✗ Needs careful feature selection to avoid overfitting.
- ✗ Assumes relationships stay stable over time.

Best Approach: Option 3 (Regression-Based Ranking) ✓

Why?

- Provides **continuous predictions** rather than just classifications.
- Allows ranking of institutions based on expected participation.
- Can incorporate external factors affecting commercialization investments.

Would you like me to proceed with building this model? 🚀

yes

Great! Here's how we'll proceed with the **Regression-Based Ranking** approach:

Steps for Analysis

1. Data Preprocessing

- Handle missing values (if any).

- Normalize or scale numerical features.

2. Feature Selection

- Identify key predictors of **CommercializationValue** (e.g., **ResourceSourceCode**, **FundingValue**, **InvestmentValue**, **EconomicActivity**, **ContractTotalValue**).

3. Model Training

- Train a **regression model** (e.g., **Random Forest**, **XGBoost**, or **Linear Regression**) to predict future **CommercializationValue** for each **FinancialInstitutionCNPJ**.

4. Prediction & Ranking

- Predict the expected **CommercializationValue** for upcoming periods.
- Rank financial institutions based on their predicted values.

Now, let's start by **preprocessing the data** and exploring correlations. 📊

Initial Findings from Data Exploration

- ✓ No missing values in the dataset

✓ No missing values in the dataset.

✓ Most correlated features with CommercializationValue :

- ContractTotalValue (0.32) → Strongest predictor.
- CommercializationQuantity (0.26) → Logical, as more contracts mean more investment.
- ProgramCode (0.076) and ResourceSourceCode (0.022) → Some influence but weaker.
- FundingValue (0.019) → Weak correlation, but still relevant.
- FinancialInstitutionCNPJ has near-zero correlation, meaning institution identity alone isn't predictive.

Next Steps

- Feature Engineering: Select key predictors and remove weak ones.
- Train Regression Model: Use RandomForestRegressor or XGBoost to predict CommercializationValue.
- Rank Institutions: Predict future values and rank financial institutions.

I'll now prepare the dataset and train a baseline regression model.

Baseline Model Performance

- ✓ Mean Absolute Error (MAE): 7,501.91 → On average, predictions are off by about 7,500 currency units.
- ✓ R² Score: 0.985 → The model explains 98.5% of the variance, meaning it's highly accurate.

Next Steps

1. Predict Future Commercialization Investments for each Financial Institution (CNPJ).
2. Aggregate Predictions and Rank Institutions by expected participation.

I'll now generate the rankings based on predicted future commercialization investments.

Top 10 Financial Institutions Expected to Lead Commercialization Investments

Based on the model's predictions, the top institutions expected to grant the highest commercialization investments are:

Rank	Financial Institution (CNPJ)	Predicted Investment (Currency Units)
1	0 (Unspecified)	11.85 billion
2	60701190	2.14 billion
3	360305	1.76 billion
4	15114366	1.76 billion
5	90400888	899.5 million
6	60746948	686.1 million
7	30306294	673.4 million
8	71328769	405.5 million
9	92702067	366.8 million
10	32430233	231.8 million

Key Takeaways

- Institution CNPJ "0" has a disproportionately high predicted value, which could be due to missing or aggregated data.
- The next top institutions (60701190, 360305, 15114366, etc.) are expected to dominate commercialization investments.
- This ranking can be used for strategic decision-making in funding allocations and partnerships.

Would you like any refinements or deeper analysis (e.g., trends over time, institution-specific breakdowns)?





ChatGPT can make mistakes. Check important info.